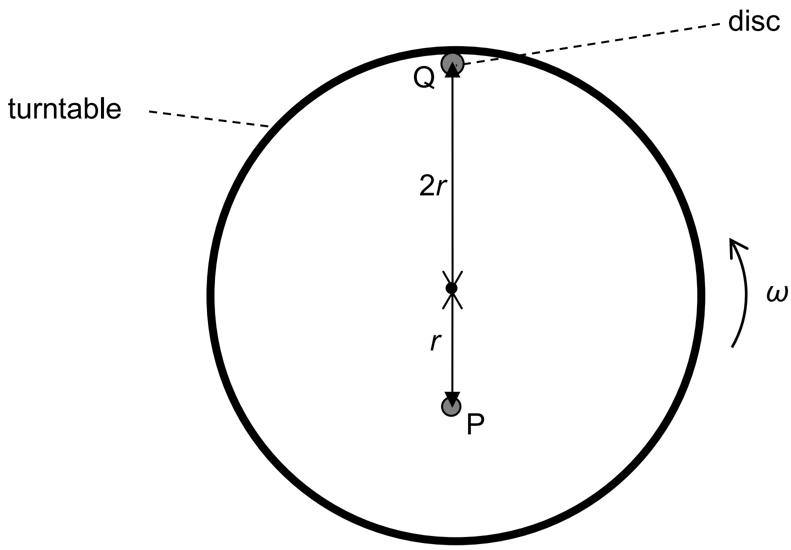
9 Two discs P and Q of mass $m$ and $2m$ respectively are placed on a rough, horizontal and level turntable as shown in the diagram. P and Q are at a distance of $r$ and $2r$ from the centre of the turntable respectively. The turntable starts rotating from rest with gradually increasing angular velocity $\omega$.




**Top View**

Given that the maximum frictional force acting on P is half of that on Q, which of the following is correct?

**A** P will slip first.

**B** Q will slip first.

**C** P and Q will slip at the same time.

**D** Neither P nor Q will slip.

Ans: B
The frictional force provides the centripetal force → $f = MR\omega^2$.
When P is about to slip, $f_{max\ P} = mr\omega_{max\ P}^2$
$\omega_{max\ P}^2 = f_{max\ P} / mr$ ----- eqn (1)

When Q is about to slip, $f_{max\ Q} = (2m)(2r)\omega^2 = 4mr\omega_{max\ Q}^2$
$\omega_{max\ Q}^2 = f_{max\ Q} / 4mr$ ----- eqn (2)

since $f_{max\ Q} = 2\, f_{max\ P}$, from eqn (2)
$\omega_{max\ Q}^2 = 2f_{max\ P} / 4mr = f_{max\ P} / 2mr$ ----- eqn (3)

Comparing eqn (1) and (3), $\omega_{max\ Q} < \omega_{max\ P}$. Hence Q will slip first.

**10** A powered spaceship is moving directly away from a planet as shown below.